



PAPER
A



2018 MATHEMATICS

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

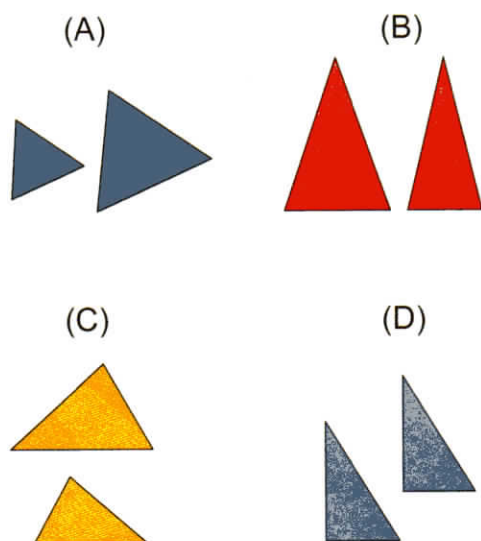
You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

ICAS

1. Mark drew two triangles of the same shape and size.

Which triangles could be the ones Mark drew?



4. Maria looks at her watch.



What time is it?

- (A) six past three
(B) half past three
(C) half past four
(D) quarter past six

2. Jess drew 2 groups of 4 butterflies.



She wants 3 groups in total.

How many more butterflies must she draw?

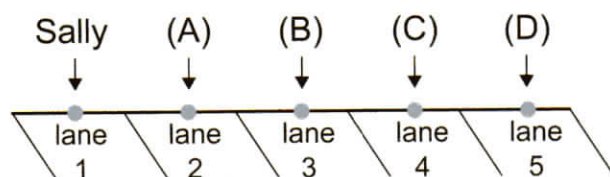
- (A) 1 (B) 4
(C) 8 (D) 12

3. In a race, four students stand in their lanes in this order:

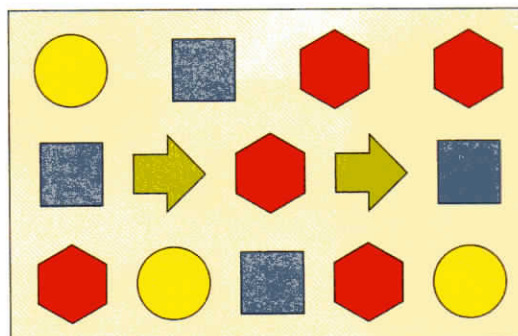
Sally, Jake, Tina, Amy.

There is one empty lane between Sally and Jake.

Where is Tina standing?



5. This mat is used for a game of Hot Spot.



A player starts by standing on any shape around the edges of the mat.

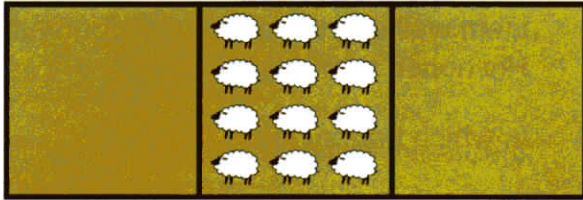
Which shape can a player **NOT** stand on at the start of the game?

- (A) (B) (C) (D)





6. Farmer Jim has three sheep paddocks.



He wants each paddock to have the same number of sheep in it.

How many sheep does he need to move in total?

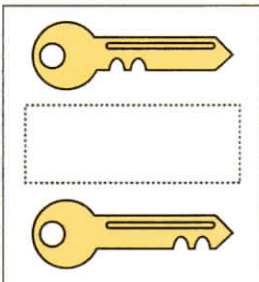
- (A) 3 (B) 4
(C) 6 (D) 8

7. Jai is using the number of cuts in the bottom edge of some keys to arrange them in order.

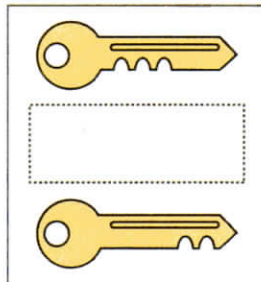


Between which two keys does this key belong?

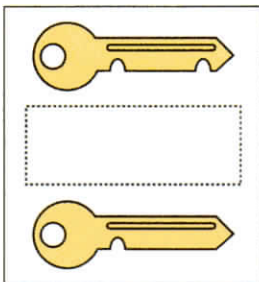
(A)



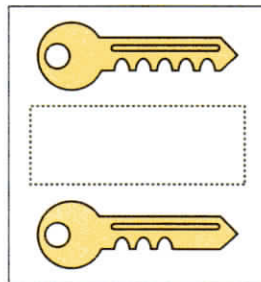
(B)



(C)



(D)



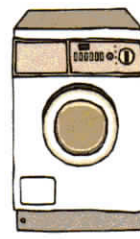
8. Which object holds about 1 litre when full?

(A)



juice carton

(B)



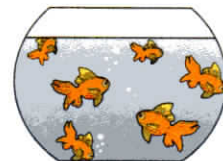
washing machine

(C)



teacup

(D)



fish bowl

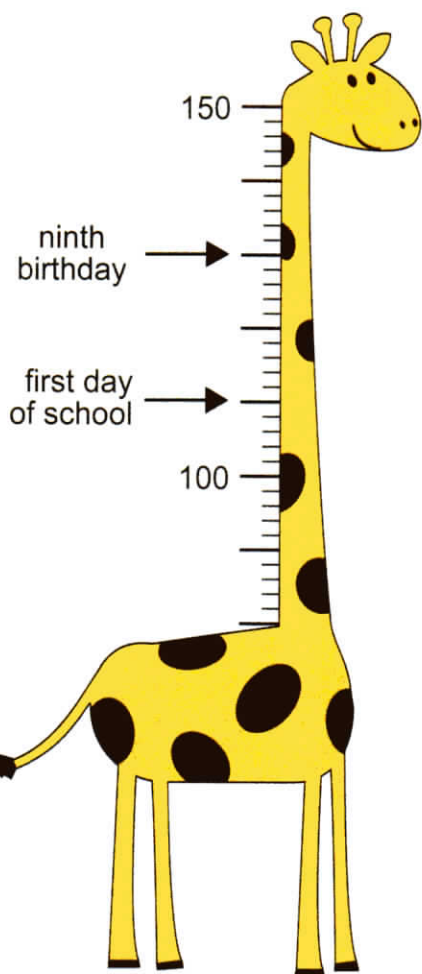
9. Susan counted 20 petals on the flowers she picked. Each flower had 5 petals.



How many flowers did she pick?

- (A) 2
(B) 3
(C) 4
(D) 5

10. The chart shows how many centimetres tall Wendy was on her first day of school and then on her ninth birthday.



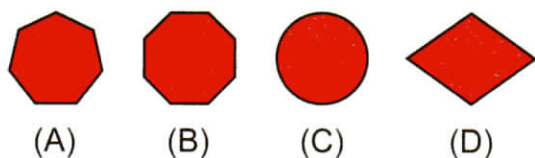
How many centimetres did Wendy grow during that time?

- | | |
|--------|--------|
| (A) 10 | (B) 20 |
| (C) 25 | (D) 30 |

11. Adam is making a pattern using shapes.

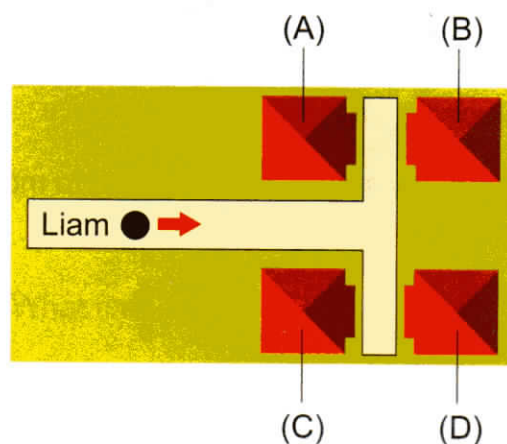


What must the next shape be in Adam's pattern?

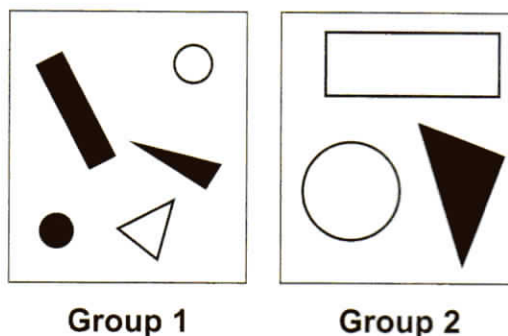


12. Liam is walking home along a footpath. He turns right onto another footpath. Liam walks to the end of this footpath. His house is on his left.

In which house does Liam live?



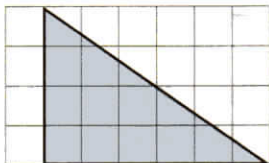
13. Priya sorted some shapes into two groups.



Which of these could be a label for one of these groups?

- | |
|------------------|
| (A) Circles |
| (B) Triangles |
| (C) Big shapes |
| (D) Black shapes |

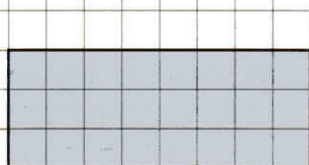
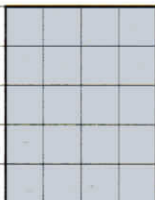
14.



Which rectangle has twice the area of this triangle?

(A)

(B)



(C)

(D)

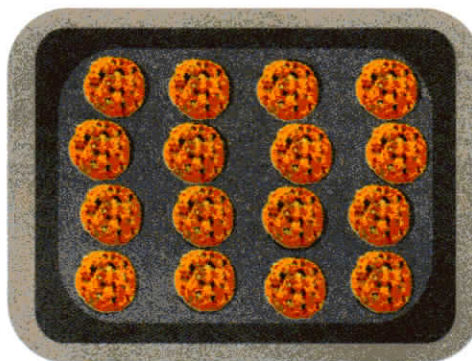
15. Rita made a card using a program on her computer. She began by drawing an image of two roses in the top left corner.



What operation did Rita use to move copies of the first image?

- (A) flip
- (B) slide
- (C) half turn
- (D) quarter turn

16. Pete baked 16 biscuits.



He ate $\frac{1}{4}$ of the biscuits. Then he gave $\frac{1}{2}$ of the remaining biscuits to his mother.

How many biscuits did Pete have left?

- (A) 4
- (B) 6
- (C) 8
- (D) 12

17. The numbers in this pattern go up by the same amount each time.

25	35	40	45			?
----	----	----	----	--	--	---

What number must ? be?

- (A) 50
- (B) 55
- (C) 60
- (D) 65

18. A mug can hold up to 15 marbles. The same mug can hold up to 5 balls.



15 marbles



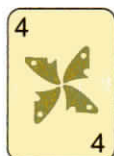
5 balls

A cake tin can hold up to 60 of these marbles.

About how many of these balls can the same cake tin hold?

- (A) 20
- (B) 40
- (C) 50
- (D) 80

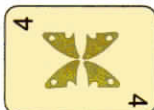
19. This is Jake's card.



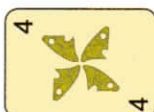
Which of these is the same card?



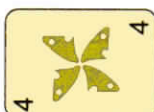
(A)



(B)

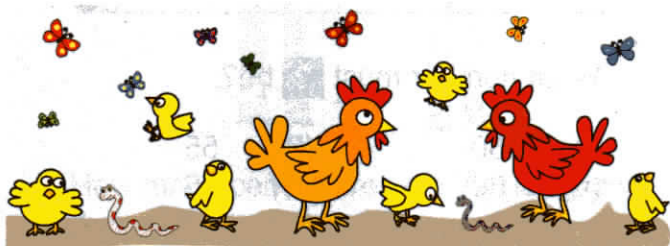


(C)



(D)

20. James is creating a table. He uses a different coloured dot to represent each chick, hen, worm and butterfly in this picture.



Which row is missing a dot?

	Animal	Number
(A)		● ● ● ● ●
(B)		● ●
(C)		● ●
(D)		● ● ● ● ● ● ● ●

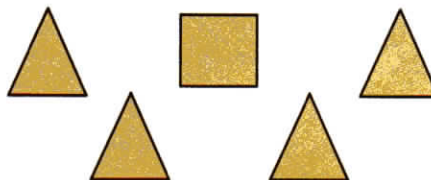
21. Jane is counting up by threes.

She starts to write this pattern in four columns as shown.

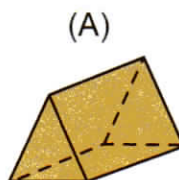
(A)	(B)	(C)	(D)
3	6	9	12
15	18	21	24
27			

In which column should Jane write 42?

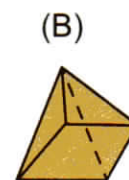
22. Arun wants to use all of these shapes as the faces of a solid.



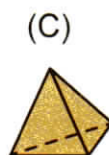
Which solid can Arun make?



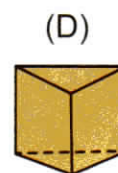
(A)



(B)



(C)

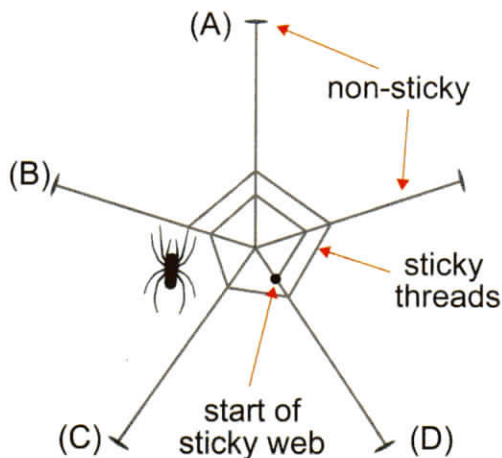


(D)

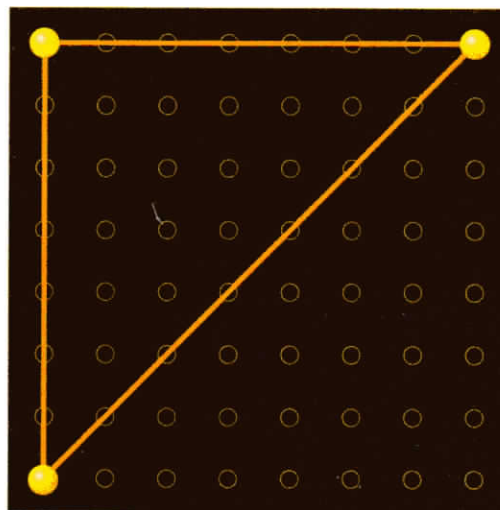
23. Charlotte the spider is spinning a web. She joins 5 non-sticky threads at the centre. Then she spins 8 sticky threads between them as shown.

Charlotte stops spinning when there are 127 sticky threads in total.

On which non-sticky thread does Charlotte stop?



25. Lin has some coloured bands and pins. She wants to make some more triangles on this pegboard with the same shape and size as this one. They can overlap but not fully.



How many more of these triangles can Lin make?

- (A) 4
(B) 3
(C) 2
(D) 1

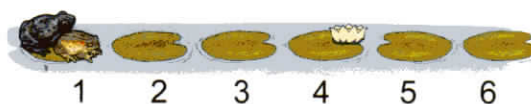
24. Alex made a table to list some of the things he planned to do on Saturday.

Activity number	Time	Activity
1	8:00 – 9:00	walk the dog
2	9:30 – 11:30	play soccer
3	11:30 – 2:00	visit a friend
4	3:00 – 3:30	piano lesson

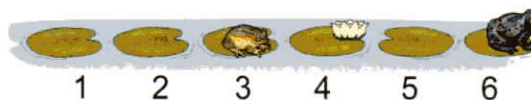
Which of these lists the activities in order from the shortest to the longest amount of time?

- (A) 1, 4, 2, 3
(B) 1, 4, 3, 2
(C) 4, 1, 2, 3
(D) 4, 1, 3, 2

26. Two frogs sit on a lily pad as shown.



The large frog jumps forward 5 lily pads at a time. The small frog jumps forward 2 lily pads at a time.



What is the number of the first lily pad that both frogs will land on?

- (A) 7
(B) 9
(C) 10
(D) 11

27. The graph shows the number of goals scored by the school basketball team over four weeks.

Week	Number of goals
1	
2	
3	
4	

 = 3 points

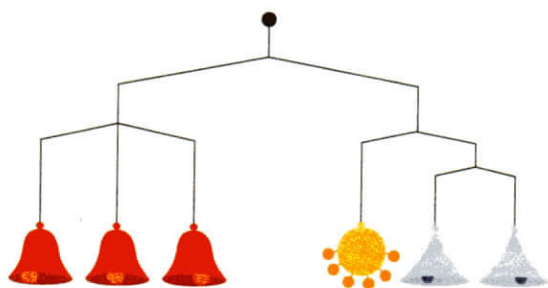
 = 2 points

 = 1 point

In which week did the team score the most points?

- (A) 1
(B) 2
(C) 3
(D) 4

28. Peta balanced 2 blue bells with a yellow bell. Then she balanced these with 3 red bells as shown.



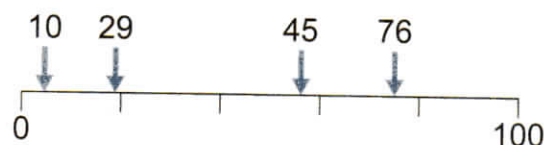
The mass of each blue bell is 6 grams.

What is the mass of one red bell?

- (A) 8 grams (B) 6 grams
(C) 4 grams (D) 2 grams

29. Max placed four numbers on a number line.

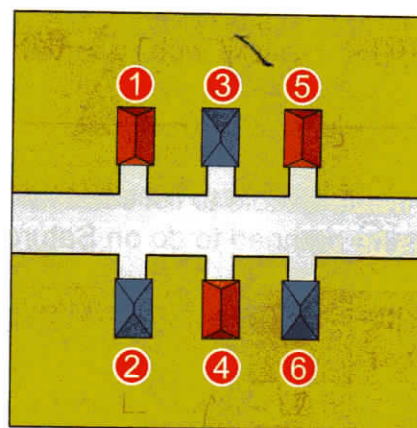
He only placed one of the numbers in the correct position.



Which number did he place correctly?

- (A) 10
(B) 29
(C) 45
(D) 76

30. Colin, Aditi, Jake, Ben, Samira and Lin live in houses on the same street.



Samira lives in house number 5. Colin lives on the same side of the street as Samira.

Aditi lives in the house between Lin and Jake. Lin lives opposite Colin.

What is the number of the house that Ben lives in?

- (A) 4
(B) 3
(C) 2
(D) 1

31. Mrs Lo is having a party for her daughter Mia.

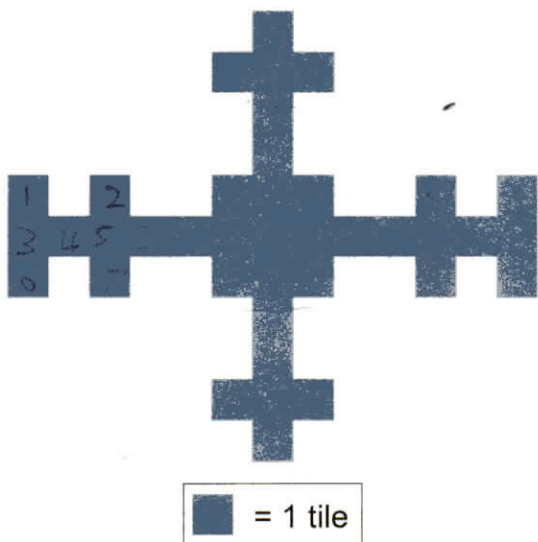
She has 25 cupcakes and 40 balloons to share equally among the children.

After Mrs Lo hands out the cupcakes, there are 4 left over. After she hands out the balloons, there are 5 left over.

How many children are at the party?

- (A) 3 (B) 5
(C) 7 (D) 9

32. Aditi joined square tiles of the same size to make this shape. The tiles do not overlap.



How many tiles did she use?

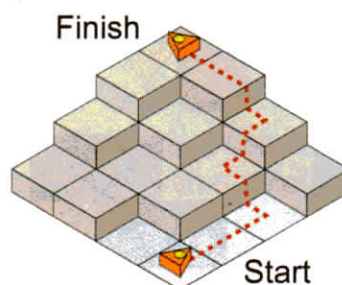
- (A) 32
(B) 33
(C) 39
(D) 41

33. In 2018 Charlie is  years old.

Which number sentence shows how old Charlie will be in 2023?

- (A)  + 23 - 18
(B)  - 23 + 18
(C)  - 18 - 23
(D)  + 18 + 23

34. Sarah is playing a computer game. The picture shows how she moved her triangle from Start to Finish on the screen.



Instruction	Key to press
Move Forward	F
Turn Left	L
Turn Right	R
Jump Up	J

The keys she pressed were:

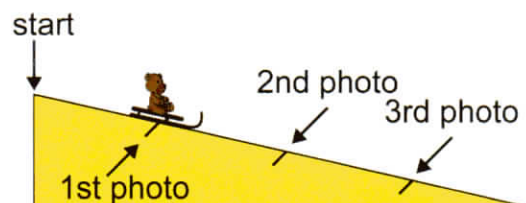
F F L J R J L J F

Which keys will move the triangle from Start to Finish along a different path?

- (A) L F R J L F R J J L F
(B) F F J L J J L F
(C) L F J R F J J F
(D) F L J F F J L J F

35. Belinda's driveway slopes so that her toy slides at the same speed from the top to the bottom.

She set up her camera to take a photo of the toy every 2 seconds after it started sliding.

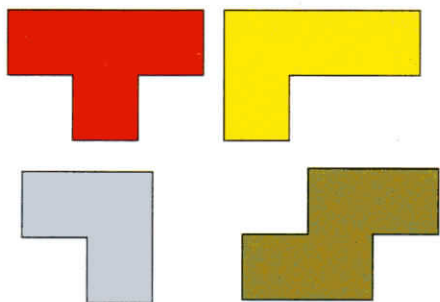


The toy slid 3 metres every second.

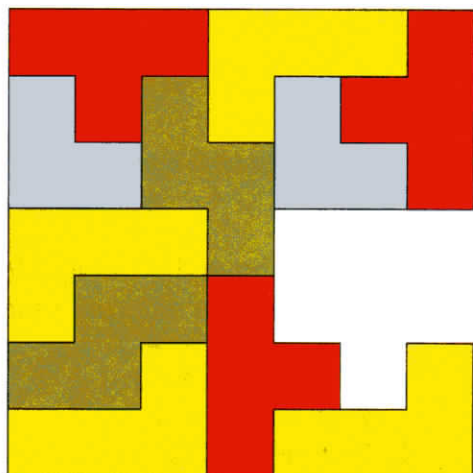
How far had the toy slid from the start when the third photo was taken?

- (A) 3 metres (B) 6 metres
(C) 12 metres (D) 18 metres

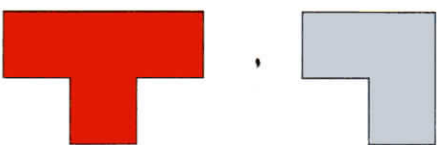
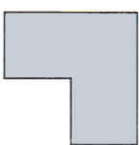
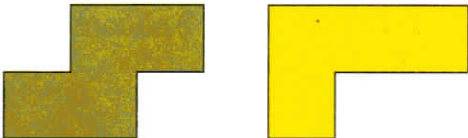
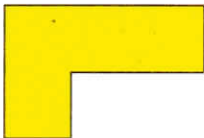
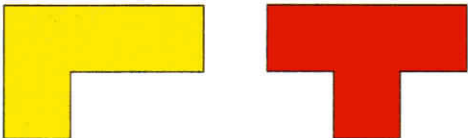
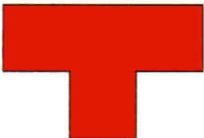
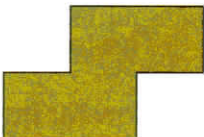
36. Zane is completing a puzzle that has only four different shapes.



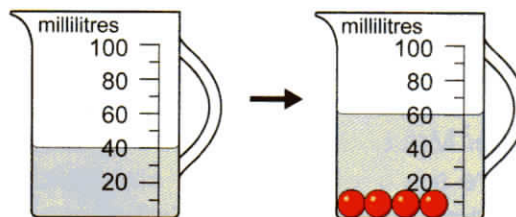
Zane needs to add two more pieces to finish the puzzle.



Which of these are the two pieces?

- (A)  , 
- (B)  , 
- (C)  , 
- (D)  , 

37. Mike dropped 4 red balls into a small measuring jug.



Then he took out half of the red balls.

What was the water level, in millilitres, in the jug after this?

- (A) 30 (B) 45
(C) 50 (D) 60

38. Natasha found a fast way to multiply two big numbers together.

For example, 38×42

	30	8
40	1200	320
2	60	16

Adding the numbers in the white boxes together gives the answer 1596.

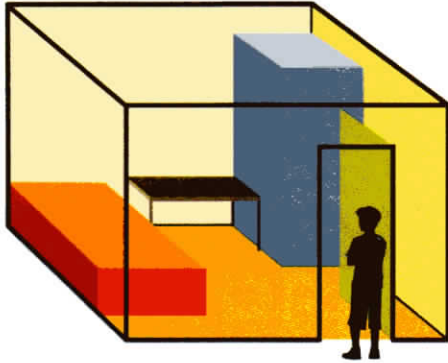
She started to use this method to solve another multiplication problem.

		5
20	800	
3		15

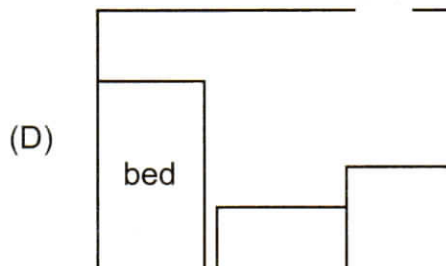
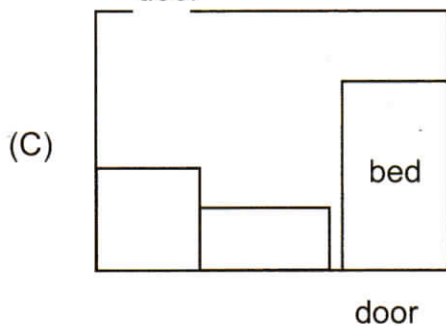
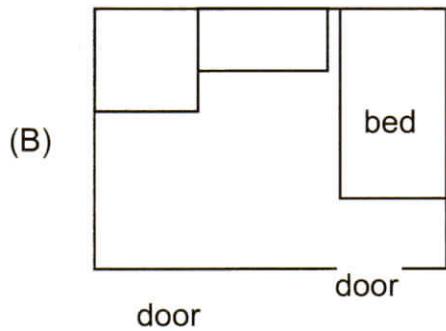
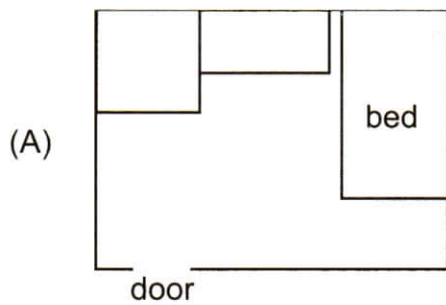
What is the answer to the problem?

- (A) 815
(B) 1035
(C) 1197
(D) 1215

39. Jamie is standing at his bedroom door looking into his room.



Which of these is a floor plan of Jamie's room?



40. Danny has 3 red socks and 4 blue socks.



How many ways can she pick two socks that are the same colour?

- (A) 5
(B) 7
(C) 9
(D) 12

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THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

Australia	Year 3
Brunei	Primary 3
Egypt	Year 3
Hong Kong	Primary 3
Indian Subcontinent¹	Class 3
Indonesia	Year 4
Malaysia	Standard 3
Middle East²	Class 3
New Zealand/Pacific³	Year 4
Singapore	Primary 2
Southern Africa⁴	Grade 3

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji.

⁴ Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.



UNSW Global

Mathematics | 2018 | Year 3

1	Space & Geometry	Identify a pair of identical triangles	D
2	Number & Arithmetic	Represent multiplication as equal groups of four	B
3	Space & Geometry	Position students according to a set of instructions	C
4	Measures & Units	Read an analogue clock	B
5	Chance & Data	Identify the shape that is not positioned along the border	A
6	Number & Arithmetic	Solve a problem involving grouping in fours	D
7	Algebra & Patterns	Find a missing element of a picture pattern that increases by ones	D
8	Measures & Units	Recognise an object with a given capacity	A
9	Number & Arithmetic	Count up by fives to a given number	C
10	Measures & Units	Calculate the difference between two heights	B
11	Algebra & Patterns	Continue a simple shape pattern	A
12	Space & Geometry	Locate a position on a map after taking left and right turns	D
13	Chance & Data	Identify the category used to sort shapes	C
14	Measures & Units	Compare the areas of 2-D shapes using informal units	D
15	Space & Geometry	Identify the type of transformation used to move an image	B
16	Number & Arithmetic	Use an array to solve a problem involving halves and quarters	B
17	Algebra & Patterns	Continue a number pattern that increases by five each time	C
18	Measures & Units	Solve a problem involving capacity	A
19	Space & Geometry	Identify a shape rotated by a quarter turn	D
20	Chance & Data	Match a picture to a table to solve a problem	A
21	Number & Arithmetic	Solve a problem with repeated addition	B
22	Space & Geometry	Identify the solid formed after joining the given faces	B
23	Number & Arithmetic	Count up by fives from a given starting point	A

24	Measures & Units	Interpret a timetable to order lapsed times	C
25	Space & Geometry	Recognise the number of identical triangles that can be made on a pegboard	B
26	Algebra & Patterns	Use multiples of 2 and 5 to solve a problem	D
27	Chance & Data	Interpret a picture graph to calculate and compare scores	D
28	Measures & Units	Determine the mass of an object on a balance scale	A
29	Number & Arithmetic	Identify the number correctly positioned on a number line	D
30	Space & Geometry	Solve a logic problem involving relative position	B
31	Number & Arithmetic	Solve a problem involving remainders	C
32	Measures & Units	Find the number of tiles used to make a composite shape	C
33	Algebra & Patterns	Identify the number sentence that can be used to solve a problem	A
34	Space & Geometry	Identify a set of directions to move from one position to another on a 3-D model	A
35	Number & Arithmetic	Solve a problem involving multiplication by two and three	D
36	Measures & Units	Identify the missing shapes based on informal area	D
37	Measures & Units	Solve a problem involving reading a scale with increments of ten	C
38	Number & Arithmetic	Solve a multiplication problem using partitioning	B
39	Space & Geometry	Interpret a map of a familiar location and identify the relative positions of objects	C
40	Chance & Data	Calculate the number of pairs that can be made from two types of objects	C